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REVIEW ARTICLE

Physiological and methodological aspects of rate of force development assessment in human skeletal muscle

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Summary

Rate of force development (RFD) refers to the ability of the neuromuscular system to increase contractile force from a low or resting level when muscle activation is performed as quickly as possible, and it is considered an important muscle strength parameter, especially for athletes in sports requiring high-speed actions. The assessment of RFD has been used for strength diagnosis, to monitor the effects of training interventions in both healthy populations and patients, discriminate high-level athletes from those of lower levels, evaluate the impairment in mechanical muscle function after acute bouts of eccentric muscle actions and estimate the degree of fatigue and recovery after acute exhausting exercise. Notably, the evaluation of RFD in human skeletal muscle is a complex task as influenced by numerous distinct methodological factors including mode of contraction, type of instruction, method used to quantify RFD, devices used for force/torque recording and ambient temperature. Another important aspect is our limited understanding of the mechanisms underpinning rapid muscle force production. Therefore, this review is primarily focused on (i) describing the main mechanical characteristics of RFD; (ii) analysing various physiological factors that influence RFD; and (iii) presenting and discussing central biomechanical and methodological factors affecting the measurement of RFD. The intention of this review is to provide more methodological and analytical coherency on the RFD concept, which may aid to clarify the thinking of coaches and sports scientists in this area.

Introduction

Rate of force development (RFD) is considered an important aspect of human skeletal muscle function *in vivo*. In fact, RFD is a key factor in rapid movements where a short contraction time may not allow maximal muscle force to be reached (Aagaard *et al.*, 2002a, 2007). Hence, a high RFD plays an important role in the ability to perform rapid and forceful movements, both in highly trained athletes and in elderly individuals who need to control unexpected perturbations in postural balance (Aagaard, 2003). In addition, RFD has been shown to be a sensitive indirect marker of acute muscle damage (Crameri *et al.*, 2007; Penailillo *et al.*, 2015; Farup *et al.*, 2016; Mackey *et al.*, 2016; Oliveira *et al.*, 2016) and exercise-induced neuromuscular fatigue (Thorlund *et al.*, 2008). Finally, RFD has also been used as an adjunctive outcome measure for return-to-sport decisions after severe knee joint injuries (Angelozzi *et al.*, 2012). For these reasons, RFD has received considerable scientific attention in recent decades. However, many of the central issues related to RFD (definition, characteristics, methodological and physiological determinants, test–retest reliability and relationship with sport performance) are often contradictory and not well understood, although recent reviews have helped to clarify the concept (Maffiuletti *et al.*, 2016).

There is little consistency between researchers in terms of the definition and rationales behind RFD. The terms 'RFD', 'explosive strength', 'rapid muscle contraction', 'rapid force capacity', 'explosive force production' and 'ballistic contraction' have often been used to describe the same concept, namely the ability of the neuromuscular system to produce a high rate of rise in muscle force per unit of time during the initial phase of rising muscle force following contraction onset. In addition, RFD appears to be highly sensitive to acute and long-term fatigue, exercise-induced muscle damage and longitudinal training, while also being affected by several methodological issues (Viitasalo & Aura, 1984; Jenkins et al., 2014; Haff et al., 2015; Penailillo et al., 2015). Moreover, different protocols and assessment methods are currently used to measure and quantify RFD. Surprisingly, this wide disparity in practice is rarely perceived as having a negative effect. Unfortunately, the disparities in methodology along with a wide interindividual variability of this parameter (Folland et al., 2014) hinder a clear and universal definition of the RFD concept. Furthermore, while the measurement of voluntary RFD is often perceived as a simple concept, it actually comprises a multifactorial measure which reflects the combined functioning of the neural and musculotendinous systems, respectively. To further complicate the concept of rapid force production, some controversy appears to exist about the physiological factors affecting RFD. Despite ample evidence of a significant role for neural and structural mechanisms in muscle contraction, only limited experimental evidence exists (Van Cutsem et al., 1998; Griffin & Cafarelli, 2005; Duchateau et al., 2006; Del Balso & Cafarelli, 2007; Folland et al., 2014) in relation to the identification of the underlying mechanisms responsible for voluntary rapid force production. Thus, many of the physiological factors proposed as being responsible for RFD appear to be speculative in view of the lack of supporting experimental data (Van Cutsem et al., 1998; Enoka & Fuglevand, 2001; Van Cutsem & Duchateau, 2005), and therefore, more extensive investigative efforts are needed within this area. Furthermore, many of these factors are modifiable through a variety of training modalities. For these reasons, RFD is an important variable not only for exercise scientists but also for coaches, athletes and clinical practitioners to understand, quantify and attempt to enhance human movement performance. Therefore, the main purpose of this article was to review the existing literature in order to (i) explore in detail the definition(s) and biomechanical characteristics of RFD; (ii) determine and discuss important neuromuscular components that influence the expression of RFD; and (iii) present and discuss important methodological aspects affecting RFD assessment.

Definition and characteristics of rate of force development

RFD is a term widely used in the scientific literature to describe the ability of the neuromuscular system to increase contractile force from a low or resting level when muscle activation is intended to be performed as quickly as possible (Schmidtbleicher, 1992; Folland et al., 2014). This parameter typically is used as a measure of the capacity for rapid force/torque production as it reflects the rate at which muscle tension can be developed (i.e. RFD = ΔForce/Δtime) (Semenick & Yates, 1989; Young & Bilby, 1993; Murphy et al., 1995). For these reasons, RFD is considered to be an important physical requirement for movements performed rapidly, especially when force production times are short (e.g. 100–300 ms), such as when reversing a fall or in various athletic events (e.g. sprints, throws and jumps). Numerous studies have indicated that RFD can be measured from the force–time or torque–time curve obtained under isometric contractions (Aagaard et al., 2002b; Suetta et al., 2004; Bojsen-Moller et al., 2005; Andersen & Aagaard, 2006; Tillin et al., 2010; Folland et al., 2014; Penailillo et al., 2015). However, RFD can also be quantified during dynamic contraction conditions (Kawamori et al., 2006; Kilduff et al., 2007; Jakobsen et al., 2012) that have greater similarities with specific sports actions (movement patterns, forces, velocities, power output and mechanical specificity) compared to isometric test conditions. Some authors (Mirkov et al., 2004) have defined the term *explosive force production* as the ability of the neuromuscular system to perform high-speed muscle actions while *RFD* refers to the testing of explosive force production. However, most studies have suggested that RFD is the most accurate, simple and unambiguous term to refer to a rapid (sudden) rise force production (Aagaard et al., 2002a; Andersen et al., 2005, 2010; Comfort, 2013; Haff et al., 2015). Further, it seems more appropriate to use the term *RFD* to refer to rapid force generation rather than *explosive strength*, as explosive strength typically is associated with high-speed dynamic movements (Zatsiorsky, 2002), whereas RFD can be measured in both static and dynamic contraction conditions. In addition, to avoid jargon a recent review has recommended that the term 'explosive' should no longer be used to describe human movement (Winter et al., 2016).

For a muscle group crossing a given joint (i.e. knee, elbow, ankle or hip), RFD is calculated *in vivo* as the slope of the moment–time curve (Δmoment/Δtime) recorded during maximal voluntary joint flexion or extension, whereas multi-joint test contractions (e.g. squat, dead lift, bench press, power clean or jumps) or in isolated muscle preparations *in vitro*, contractile RFD is defined as the slope of the force–time curve (i.e. Δforce/Δtime) (Schmidtbleicher, 1992; Hakkinen et al., 1998; Aagaard et al., 2002a; Suetta et al., 2004; Tillin et al., 2010; Folland et al., 2014; Waugh et al., 2014; Penailillo et al., 2015). This parameter can be expressed in absolute terms (N s^{-1}) or relative to peak force (to yield *relative RFD*), body mass (BM) or muscle cross-sectional area (CSA) (Aagaard et al., 2002b; Suetta et al., 2004; Waugh et al., 2013). Considering that RFD is determined and expressed by the slope of the force–time curve (Schmidtbleicher, 1992; Aagaard et al., 2002a), it can be measured from the onset of contraction to any point of the force–time (or moment–time) curve or between any two points on the curve (Tillin et al., 2010; Buckthorpe et al., 2012). Therefore, a subject can have as many RFD values as the number of time intervals within the force–time curve that are captured in the measurement process. In this regard, it is very important to report the sampling time bands used when calculating specific RFD values because this factor directly will influence the absolute and relative RFD values obtained from the force/torque–time curve (Aagaard et al., 2002a; Andersen et al., 2010; Haff et al., 2015). From a practical perspective, the time interval chosen to calculate RFD will depend on the aim of the measure, including assessment of the strength qualities of athletes, monitoring training effects on fatigue or muscular performance, and including the analysis of sports-specific techniques. Several studies (Andersen *et al.*, 2010; Oliveira *et al.*, 2013; Waugh *et al.*, 2013) have indicated that analysing RFD over a wide range of time intervals seems of importance for reaching a more full interpretation of the results because: (i) different sporting movements and daily tasks require forces to be produced within different time frames; and (ii) RFD calculated in different time intervals from the onset of contraction may be affected by different physiological mechanisms/parameters and therefore could respond differently in response to resistance training and/or acute fatigue (Andersen *et al.*, 2005, 2010; Barry *et al.*, 2005; Andersen & Aagaard, 2006; de Oliveira *et al.*, 2013; Oliveira *et al.*, 2013; Waugh *et al.*, 2013). Thus, the evaluation of training-induced changes in RFD over different time epochs following onset of contraction is of significant interest to scientists as well as strength and conditioning researchers and coaches because it may provide insight into the physical condition of patients and athletes and may also provide important information about the effect(s) of neuromuscular training (Marshall *et al.*, 2011).

Maximal RFD (MRFD) is typically quantified as the peak slope of the force–time curve (Aagaard *et al.*, 2002b; Suetta *et al.*, 2004; Haff *et al.*, 2005, 2015; Buckthorpe *et al.*, 2012). The time period over which the peak (maximal) rate of change in force has been determined has varied from a moving-average interval of 2 ms up to 60 ms (Viitasalo *et al.*, 1980; Christ *et al.*, 1993, 1994; Wilson *et al.*, 1993; Aagaard *et al.*, 2002b; Andersen & Aagaard, 2006; Andersen *et al.*, 2010; Beckham *et al.*, 2013; Haff *et al.*, 2015). The available data indicate that increasing the time interval from 2 to 60 ms reduces the magnitude of the MRFD (Wilson & Murphy, 1996; Haff *et al.*, 2015). However, the effects on reliability and validity are controversial. Wilson *et al.* (1996) indicated that the use of wider time intervals did not appreciably affect test–retest reliability or validity of MRFD, whereas a recent study performed by Haff *et al.* (2015) conversely showed that the width of the analysis interval used can impact on reliability values. In this study (Haff *et al.*, 2015), peak RFD was calculated with the use of 2, 5, 10, 20, 30 and 50 ms sampling windows and showed that calculating MRFD using 20-ms moving sampling intervals resulted in a greater absolute reliability (CV: 12·9%) compared to MRFD calculated using shorter and longer time intervals (CV: ~15–19%). Based on these results, Haff *et al.* (2015) recommended a moving sampling window of 20 ms to be used for the determination of peak RFD.

Unlike other parameters such as the peak force (or torque), which is typically achieved at about 400–600 ms after the onset of contraction during isometric activations (Sukop *et al.*, 1974; Thorstensson *et al.*, 1976a, 1976b; Zatsiorsky, 1995; Aagaard *et al.*, 2002a) or 150–400 ms (depending on the relative intensity and exercise used) during dynamic activations (Kawamori *et al.*, 2006; Jimenez-Reyes *et al.*, 2016), MRFD measured through both an isometric test or during the static phase of a dynamic test is usually attained within the first 100 ms after the onset of contraction (Fig. 1) (Kearney & Stull, 1981; Christ *et al.*, 1993; Haff *et al.*, 1997; Aagaard *et al.*, 2002a; de Ruiter *et al.*, 2004, 2007, 2010; Gruber *et al.*, 2007; Molina & Denadai, 2012). On the other hand, studies that have investigated the kinematics and kinetics associated with different external loads (Schmidtbleicher, 1992; Kawamori *et al.*, 2005, 2006; Kilduff *et al.*, 2007; Cormie *et al.*, 2008; Stone *et al.*, 2008; Comfort *et al.*, 2011, 2012) have indicated that MRFD is similar for all loads lifted from 30% to 100% of 1RM (Fig. 2). This means that at muscle loadings higher than 25–30% of maximal dynamic strength, MRFD is likely to be stable and to converge towards its maximum (peak) value in absolute terms, whereas with loads lower than 25% of 1RM, MRFD may not be reached (Schmidtbleicher & Haralambie, 1981; Schmidtbleicher, 1992). This is due to loads lower than 25–30% of 1RM providing a low resistance; consequently, the load would move before the force needed to produce MRFD has been applied. In addition, MRFD has another special characteristic. During isometric activations, the force or torque level obtained at the instant when MRFD is reached corresponds to approximately 30% of maximal (MVC) force

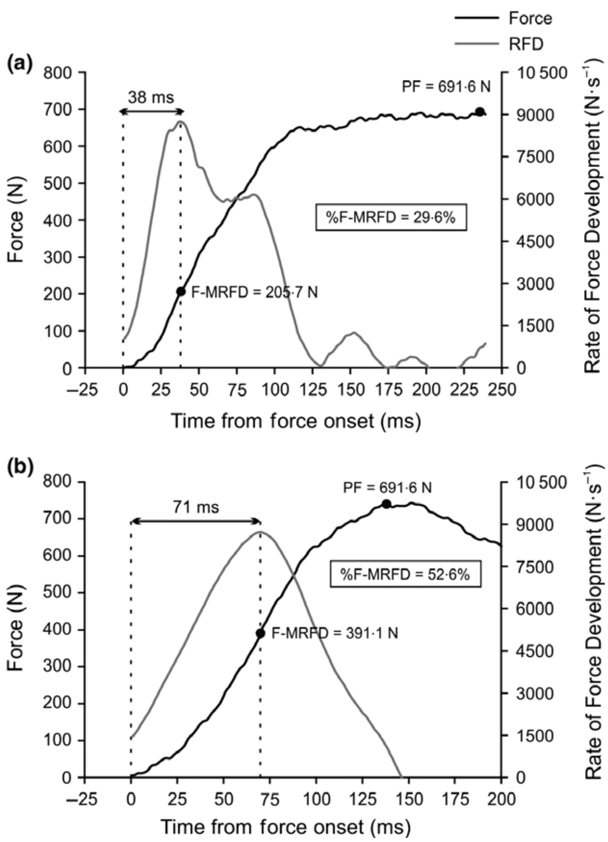

Figure 1 Force–time curve during (a) isometric and (b) dynamic (50% 1RM) bench press exercise. The maximal rate of force development (MRFD) was determined as the highest rate of force development during a 50-ms sampling windows. In both types of contractions, MRFD is achieved before 100 ms. The percentage of peak force at MRFD (%F-MRFD) during isometric and dynamic activations were 29·6% and 52·6%, respectively.

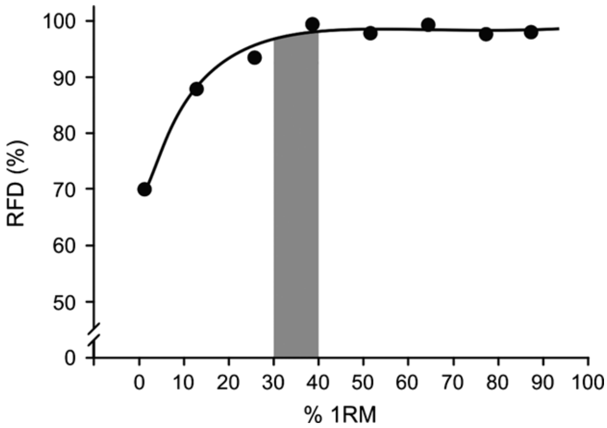

Figure 2 Representation of relationship between maximal rate of force development (%) and relative intensity (% 1RM) during bench press exercise (*Unpublished data*).

production (Fig. 1a). During dynamic contraction conditions such as concentric bench press exercise, this variable may be recorded between 40% and 60% of peak force achieved in such muscle activation, depending on the relative intensity used (Rodríguez-Rosell, D., Yáñez-García, J. M., Mora-Custodio, R. González-Badillo, J. J., *unpublished data*). This variable (percentage of force at MRFD) has received little scientific attention to date, despite that this parameter might be decisive for performance in various high-speed movement actions (Demura *et al.*, 2003; Marques *et al.*, 2011).

Physiological factors that can affect rate of force development

Muscle contraction is recognized as a complex and multifactorial process that depends on both neural and structural factors (Bell & Jacobs, 1986; Aagaard, 2003; Folland & Williams, 2007). Both the magnitude and speed of muscle shortening play a key role in determining the maximal force and rate of force rise at each instant of the contraction phase (Desmedt & Godaux, 1977; Ricard *et al.*, 2005). Several studies (Kyrolainen & Komi, 1994; Pryor *et al.*, 1994; Cormie *et al.*, 2009; Tillin *et al.*, 2010; Ereline *et al.*, 2011; Folland *et al.*, 2014) have concluded that rapid force production differs markedly between individuals even in homogeneous cohorts, which suggests that the underlying mechanism(s) of rapid muscle contractions may differ substantially between individuals. Thus, this section seeks to summarize the most relevant neuromuscular factors affecting the expression of RFD. For a more detailed description of the neural and muscular determinants of RFD and the neuromuscular adaptations to training that influence RFD, we refer the reader to recent review by Maffiuletti and co-workers (Maffiuletti *et al.*, 2016).

Neural factors

Motor unit discharge rate

Motor unit (MU) discharge rate seems to be one of the most important factors influencing contractile RFD (Sale, 1988; Van Cutsem *et al.*, 1998; Gabriel *et al.*, 2001, 2006; Aagaard *et al.*, 2002a, 2002b; Aagaard, 2003; Duchateau *et al.*, 2006; Holtermann *et al.*, 2007; Duchateau & Baudry, 2014). Studies analysing intramuscular myoelectrical activity during rapid muscle contractions (Van Cutsem *et al.*, 1998; Duchateau & Baudry, 2014) have indicated higher MU discharge rates at the onset of maximal voluntary muscle contraction (~200 Hz) where MRFD typically is obtained, compared to much lower MU discharge rates at the instant of maximal force generation (15–35 Hz). Other studies have demonstrated that RFD continues to increase at stimulation rates higher than the rate producing maximum tetanic tension (Miller *et al.*, 1981; Sale, 1992; Nelson, 1996; de Haan, 1998; Van Cutsem *et al.*, 1998; Duchateau *et al.*, 2006). In fact, it has been shown that MRFD can only be obtained using very high pulse rates during electrical muscle stimulation (Buller & Lewis, 1965; Nelson, 1996; de Haan, 1998). These pulse rates are about double those needed for eliciting maximal isometric force production *in vivo* (Sale, 1992; de Ruiter *et al.*, 1999). Therefore, it is possible that such supramaximal MU discharge rates in the initial phase of a muscle contraction serve to maximize RFD rather than to influence maximal contraction force *per se* (Burke *et al.*, 1976; Behm, 1995; Aagaard, 2003; de Ruiter *et al.*, 2004; Duchateau *et al.*, 2006). Taken together, these observations suggest that maximal motor unit discharge rate plays a critical role in the ability to achieve rapid force development during the initial phase of rising muscle force (de Ruiter *et al.*, 2004; Duchateau & Baudry, 2014).

Doublet discharges

RFD also appears to be influenced by so-called doublet discharge or doublet behaviour of active MUs (Van Cutsem *et al.*, 1998; Aagaard, 2003; Christie & Kamen, 2006; Mrowczynski *et al.*, 2015). A doublet discharge has been defined as a pair of action potentials with an interspike interval below 5 ms (Esslen *et al.*, 1969; Van Cutsem *et al.*, 1998; Mrowczynski *et al.*, 2015), which likely arise as a result of delayed depolarization in the dendrites of spinal motoneurons (Calvin & Schwindt, 1972). Although their function is not completely understood, it seems that the doublets induce increased sarcoplasmic reticulum Ca^{2+} release from the sarcoplasmic reticulum at the onset of contractions, which increases peak force and force–time integral (Binder-Macleod & Kesar, 2005; Cheng *et al.*, 2013). For this reason, the occurrence of doublets is considered an important adaptive mechanism that can increase both the rate and amount of force production (Van Cutsem *et al.*, 1998; Aagaard, 2003; Griffin & Cafarelli, 2005; Van Cutsem & Duchateau, 2005; Christie & Kamen, 2006; Gabriel *et al.*, 2006; Cheng *et al.*, 2013; Mrowczynski *et al.*, 2015). Previous studies (Bemben *et al.*, 1991; Izquierdo *et al.*, 1999a; Klass *et al.*, 2008) have shown that with increasing age, a progressive decline in the RFD occurs, which is accompanied by a reduced incidence of doublet discharge behaviour at the onset of contraction, presumably contributing to the reduction in RFD with increasing age (Christie & Kamen, 2006; Klass et al., 2008). Similarly, it has been found that the occurrence of doublet discharges increases with increasing muscle contraction velocity (Griffin et al., 1998; Van Cutsem et al., 1998) and with muscle fatigue to maintain a sustained level of contractile force production (Binder-Macleod & Barker, 1991; Cheng et al., 2013; Mrowczynski et al., 2015). These findings suggest that MU doublet discharge behaviour represents a unique strategy of the central nervous system to improve the efficiency of motor tasks requiring large and fast force generation, especially during the early phase of muscle contraction (Kudina & Andreeva, 2013; Mrowczynski et al., 2015).

Synchronization of motor units

The synchronization of motor units refers to the level of correlation between the timing of the action potentials discharged by concurrently active motor units (Duchateau et al., 2006; Folland & Williams, 2007). Although not affecting maximal force *per se* (for review, see Duchateau et al., 2006), this factor may optimize the rate of force generation via increased superposition of motor unit force twitches (Celichowski, 2000; Semmler, 2002; Ricard et al., 2005; Tillin et al., 2010). Supporting this notion, previous studies (Milner-Brown et al., 1975; Semmler & Nordstrom, 1998; Semmler, 2002; Folland & Williams, 2007) have reported increased MU synchronization in strength athletes compared to endurance athletes or untrained individuals, which has been accompanied by similar observations for RFD (Kyrolainen & Komi, 1994; Paasuke et al., 2001a; Andersen et al., 2007). However, in spite of its apparent wide acceptance, there is a lack of empirical evidence supporting the notion that increased common synaptic inputs to spinal motor neurons result in enhanced MU synchronization under voluntary conditions (Kline & De Luca, 2016). In fact, a recent study (Kline & De Luca, 2016) concluded that as the force generated by the muscle increases, the firing rate slope decreases and the synchronization consequently decreases. Consequently, whether MU synchronization is an optimizing factor for RFD remains unclear.

Other neural factors

Using H-reflex recording, it has been observed that resistance training induces an increase in the excitability of spinal motoneurons during isometric maximal voluntary muscle contractions (Aagaard et al., 2002b; Aagaard, 2003; Duclay et al., 2008). Although less well studied, this elevation in spinal motoneuron excitability appears to be associated with several factors, potentially including a reduction in motor unit recruitment thresholds and an increased magnitude of efferent neural drive to active myofibres, estimated by elevated V-wave responses (Aagaard et al., 2002b; Del Balso & Cafarelli, 2007; Vila-Cha et al., 2012). The observed increase in V-wave amplitude (normalized to Mmax) might also reflect, at least in part, decreased presynaptic inhibition of spinal motor neurons and/or downregulation of postsynaptic inhibitory pathways (Hamada et al., 2000; Aagaard et al., 2002b; Aagaard, 2003; Cormie et al., 2007; Holtermann et al., 2007; Andersen et al., 2009), which altogether could be involved in the observed training-induced increase in RFD. In fact, Holtermann et al. (2007) observed a positive association ($r = 0{\cdot}56$; $P<0{\cdot}05$) between gains in H-reflex excitability (recorded at 20% MVC) and RFD measured during MVC efforts for the plantar flexors following 3 weeks of isometric resistance training (Holtermann et al., 2007). Despite these findings, further research appears to be needed to uncover the specific influence of these neural factors on the training-induced gain in RFD during the initial phase of rising muscle force.

Structural factors

Fibre-type distribution and myosin heavy chain isoforms

Early reviews (Fitts & Widrick, 1996) reported that maximal cross-bridge cycle transition rate appears to be a major limiting factor for maximal intrinsic RFD of mammalian muscle fibres. Notably, maximum shortening velocity and force output of type IIA and particularly type IIX muscle fibres markedly exceed that of type I muscle fibres (Bottinelli et al., 1996, 1999), which may be especially important for mechanical muscle output in the very early phase of contraction. For this reason, although it is well established that a given histochemically typed human fibre may contain more than a single type of MHC isoform (Biral et al., 1988; Bottinelli et al., 1996; Harridge et al., 1996; Williamson et al., 2001), myofibre phenotype is often considered one of the major factors impacting muscular RFD (Maffiuletti et al., 2016). This is in line with several studies, which indicate that muscles as well as isolated fibres containing high proportions of MHC-IIX and/or MHC-IIA are characterized by high intrinsic RFD (Buchthal & Schmalbruch, 1970; Mero et al., 1981; Larsson & Moss, 1993; Harridge et al., 1996; Harridge, 2007). Thus, contractile RFD could be largely dependent on the ratio of type II versus I MHC isoforms (Harridge et al., 1996; Aagaard & Andersen, 1998; Korhonen et al., 2006). In this regard, several studies have reported positive correlations to exist between MRFD and type IIX fibre area percentage (Viitasalo & Komi, 1978; Viitasalo et al., 1981; Hakkinen et al., 1984; Harridge et al., 1996; Korhonen et al., 2006; Farup et al., 2014). In addition, parallel decrements ($r = 0{\cdot}61$; $P<0{\cdot}05$) in the proportion of type IIX fibres and relative RFD (RFD normalized to MVC) have been observed after resistance training programmes using heavy loads (Andersen et al., 2010). Collectively, these observations suggest that a general association may exist between RFD and fibre-type composition. However, Schilling et al. (2005) found that the variance in RFD explained by the percentage of MHC-IIX expression was only 2·7%. Furthermore, other studies have reported significant increases in RFD following 8–21 weeks of strength training while the proportion of type IIX fibres remained unchanged or decreased (Hakkinen et al., 2003; Winchester et al., 2008; Aagaard et al., 2011), indicating that factors other than fibre-type composition (i.e. muscle CSA, neural drive) probably play an (equally) important role for RFD. These conflicting findings underline that further research is needed to determine the role of muscle fibre composition on the expression of RFD in vivo.

Myofibre cross-sectional area and anatomical muscle size

In addition to fibre-type composition (MHC isoform content), another important factor affecting RFD is the CSA of muscle (Aagaard et al., 2003; Suetta et al., 2004) and fibres (Harridge et al., 1996). Several studies found a positive relationship between the CSA of the quadriceps femoris muscle group and RFD of the knee extensor (Izquierdo et al., 1999a), whereas other studies (Hakkinen et al., 2003; Ronnestad et al., 2012) have shown concurrent increments in quadriceps CSA and RFD measured during isometric leg extension after a 12–21 weeks of strength training period. With regard to myofibre CSA, some studies (Aagaard et al., 2007) found that old master athletes (68–78 years) with elevated type IIA and IIX muscle fibre CSA were able to produce greater absolute RFD and relative RFD compared to master athletes with lower type II myofibre CSA and age-matched untrained individuals, respectively. Yet other studies have observed parallel increments in both type IIA and IIX muscle fibre CSA and RFD following 14–21 weeks of resistance training (Hakkinen et al., 2003; Andersen et al., 2010). Thus, it is apparent that training aimed to increase the CSA of muscle and in particular of fast-twitch type II myofibres will be particularly effective in eliciting gains in RFD.

Tendon structure

Muscle tendons consist of connective tissue fibres that are extensible but without active contractile properties, whose function is the transmission of contractile forces from muscle to bone (Reeves et al., 2003). The time course of force development is affected by the tendon mechanical properties (Wilkie, 1949; Butler et al., 1978; Dunn & Silver, 1983; Bojsen-Moller et al., 2005). Thus, tendon stiffness affects the time required to stretch the series elastic component and therefore affects RFD in both adults and children (Reeves et al., 2003; Muraoka et al., 2004; Bojsen-Moller et al., 2005; Waugh et al., 2013, 2014). A greater stiffness of the tendinous structures would allow a more effective force transmission from the contractile elements to the bone, resulting in an increase in RFD (Waugh et al., 2013, 2014). In contrast, a less stiff tendon would mean that more time would be required to develop force (Burgess et al., 2007). In fact, during measurement of the applied force in an isometric action, it has been observed that RFD decreases when a compliant structure is inserted between the subject and the force transducer (Wilkie, 1949). Furthermore, MRFD is positively correlated with muscle aponeurosis–tendon stiffness (Reeves et al., 2003; Bojsen-Moller et al., 2005; Waugh et al., 2013), while concurrent gains in tendon stiffness and RFD have been reported after strength training programmes (Reeves et al., 2003; Burgess et al., 2007), acute passive stretching (Ce et al., 2008) and 20 days bed rest (Kubo et al., 2000). In terms of factors of importance for tendon stiffness, it appears that tendon thickness and length can also influence the ability to rapidly apply contractile force, as indicated by linear relationships between these variables and RFD recorded during the early phase of force production in CMJ, SJ and DJ testing (Earp et al., 2011). Taken together, these observations have prompted the suggestion that the stiffness of passive tissue structures responsible for the transmission of contractile forces from muscle to bone (mainly tendons and aponeuroses) is likely to affect the capacity for rapid increases in force at the onset of contraction. In support of this notion, the mechanical properties (stiffness) of deeply located intramuscular aponeurosis tissue may account for up to ~30% of the variance in isometric RFD when assessed during maximal knee extensor contraction (Bojsen-Moller et al., 2005).

Muscle geometry

Muscle fibre architecture (e.g. muscle fibre pennation and fascicle length) is considered an important determinant of a muscle's mechanical function (Blazevich, 2006; Azizi et al., 2008; Earp et al., 2011). Muscle fibre pennation angle refers to the angle at which muscle fascicles are oriented relative to the tendon or aponeurosis to which they attach, and it is believed to contribute to an enhanced force production capability by allowing for increased physiological muscle fibre CSA for a given anatomical muscle volume or whole muscle CSA (Kawakami et al., 2000; Aagaard et al., 2001; Blazevich, 2006; Blazevich et al., 2006). Although the effective muscle fibre shortening speed (i.e. the effective projection of the fibre speed vectors onto the origin-insertion line of pull of the whole muscle) inevitably will decrease at more steep fibre pennation angles, the fibre rotation that occurs during ongoing concentric contraction conversely contributes to increase the origin-insertion velocity of the whole muscle (Gans, 1982; Azizi et al., 2008; Maffiuletti et al., 2016) by allowing the muscle to function at a higher gear ratio (elevated muscle origin-insertion velocity/fibre velocity) (Gans, 1982; Azizi et al., 2008), which potentially may increase RFD for a given overall muscle shortening speed compared to a non-pennate muscle. However, to the best of our knowledge, only few studies (Earp et al., 2011; Erskine et al., 2014) have analysed the impact of pennation angle on RFD in human skeletal muscle. Erskine et al. (2014) showed that the percentage of change in muscle fascicle pennation angle was inversely correlated with percentage of change in normalized force at 150 ms after force onset ($r = -0{\cdot}362$; $P<0{\cdot}05$) following 12-week resistance training for the elbow flexors, indicating that an increase in muscle fascicle pennation angle produces a decrease in the shortening velocity of the whole muscle, accompanied by a reduction in RFD (Erskine et al., 2014). On the other hand, Earp et al. (2011) reported that vastus lateralis ($r = 0{\cdot}435$, $P = 0{\cdot}03$) and gastrocnemius ($r = 0{\cdot}434$, $P = 0{\cdot}03$) fascicle pennation angles were positively associated with RFD produced during DJ testing in the very early (0–10 ms) time interval from onset of contraction whereas no such relationship could be found during CMJ or SJ testing. Therefore, further studies are required to fully explore the influence of muscle fascicle pennation angle on RFD with different modes and varying intensities of muscle activation.

On the other hand, fascicle length may be assumed to reflect the total number of sarcomeres in series. An increment in the number of in-series sarcomeres should result in an increased fibre shortening speed for a given sarcomere shortening speed. This would allow for a more rapid stretching of passive series elastic structures of the muscle–tendon unit, including the distal tendon, and thus potentially lead to increased early-phase RFD (Wilkie, 1949; Blazevich et al., 2009). However, studies analysing the influence of fascicle length on RFD have shown contradictory results. For example, Earp et al. (2011) found positive correlations between RFD (0–10 and 10–30 ms) measured during CMJ testing and gastrocnemius fascicle length ($r = 0{\cdot}461$–$0{\cdot}476$; $P<0{\cdot}05$), whereas other studies (Alegre et al., 2006) have reported concurrent increases in RFD and fascicle length, respectively, in response to 13 weeks of dynamic resistance training. In contrast, a negative correlation ($r = -0{\cdot}485$, $P = 0{\cdot}014$) between gastrocnemius fascicle length and RFD measured during DJ testing also has been reported (Earp et al., 2011). Furthermore, Blazevich et al. (2009) reported training-induced changes in the fascicle length of the quadriceps muscle, which were inversely related ($r = -0{\cdot}66$ to $0{\cdot}71$; $P<0{\cdot}05$) to changes in contractile RFD measured during the very initial phase of muscle contraction (0–30 ms). These results suggest that training-induced increases in muscle fascicle length may in some situations (when shifts in force–length properties are also evoked or/and when RFD is tested on the ascending limb of the force–length relationship) lead to a reduction or complete absence of adaptive gains in contractile RFD when measured at certain joint angles, especially in the very initial contraction phase (Blazevich et al., 2009). Therefore, although several authors have pointed that elongated muscle fascicle lengths could contribute to improved RFD and increased contractile power production, other investigations have indicated that there may be a conflict or potential limit to the effect of these factors on RFD measured *in vivo* (Blazevich et al., 2006, 2008, 2009; Winchester et al., 2008). According to Blazevich et al. (2009), this may stem from the following factors: (i) human joints are typically spanned by several muscles, and it is not possible to get an accurate representation of fibre or fascicle lengths in these complex muscle groups; and (ii) some form of intervention is required to change fascicle length that does not also simultaneously result in substantial changes in other parameters, including the rate of muscle activation. Thus, it remains debatable whether fascicle length and RFD are in fact uniformly related, especially for late-phase RFD (100–250 ms), which have relevance for some sports events.

Sarcoplasmic reticulum function

Intrinsic changes in sarcoplasmic reticulum calcium sequestration function also can be implicated in the ability to apply a steep rise in contractile force development (Wahr & Rall, 1997; Rall & Wahr, 1998). It has previously been reported (Metzger & Moss, 1990) that myofibres with high sensitivity to calcium activation produced greater RFD than myofibres with low calcium sensitivity. Thus, an increased calcium sensitivity of the contractile apparatus after strength training is one of the factors that could contribute to the training-induced gain in RFD (Godard et al., 2002). Likewise, an increase in the rate of release of calcium from the sarcoplasmic reticulum would cause a faster increase in the concentration of free intracellular calcium, hence allowing the fraction of bound cross-bridges to increase more rapidly and therefore potentially allowing myofibrillar force production to develop at a faster rate (Ortenblad et al., 2000a, 2000b; Tupling et al., 2000; Binder-Macleod & Kesar, 2005; de Ruiter et al., 2006; Cheng et al., 2013).

Summary

RFD appears to be influenced by a number of neural, morphological and structural factors (Fig. 3). However, only few previous studies have addressed the relative importance of neural and contractile characteristics in determining rapid force production (Maffiuletti et al., 2016). It seems evident that the relative contribution of neural versus intrinsic contractile properties may vary throughout the time course of rise in the force–time curve (Andersen & Aagaard, 2006; Andersen et al., 2010). In this regard, it seems that early phase of RFD is dominantly influenced by neural drive and intrinsic muscle properties (Van Cutsem et al., 1998; Gruber & Gollhofer, 2004; Andersen & Aagaard, 2006; Andersen et al., 2010; Folland et al., 2014), whereas late phase of RFD is more closely affected by maximal muscle strength (Andersen et al., 2010; Folland et al., 2014) although concurrently also influenced by neural drive and peripheral muscle properties (Suetta et al., 2004; Bojsen-Moller et al., 2005; Andersen & Aagaard, 2006). Tangential RFD measured at a specific point in time probably reflects the combined influence of contributory variables prior to and around that instant, whereas sequential RFD assessed over discrete consecutive time periods may more effectively isolate any changing contributions of neural and contractile determinants throughout the time course of contraction (Folland et al., 2014). For example, after the early phase of contraction (i.e. >100 ms), neural activation may be high/maximal in all individuals, and the physiological determinants of late-phase RFD may therefore be more dependent upon muscular factors such as anatomical muscle CSA. Notably,

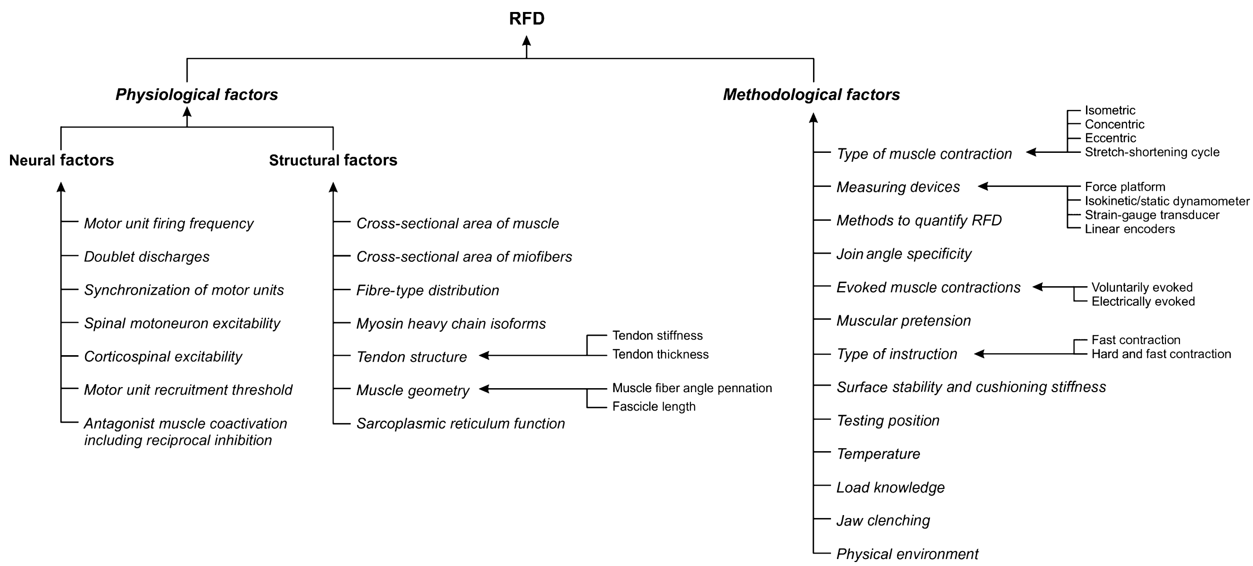

Figure 3 Flow diagram showing that RFD can be affected by several physiological and methodological factors.

however, sequential RFD obtained during later time periods may also be influenced by the force reached and/or time history of contraction prior to that time point (Tillin *et al.*, 2010).

Methodological considerations in rate of force development assessment

Analytic methods to quantify rate of force development

Probably the most important factor impacting on the absolute RFD value, its reliability and its relationship with athletic performance is the specific method employed for analysing RFD from the force–time curve (Wilson *et al.*, 1995; Tillin *et al.*, 2013; Haff *et al.*, 2015). Currently, a number of different methods have been reported in the literature that can be used for calculating both absolute RFD and relative RFD from force–time curves (Fig. 4). In a majority of these studies (Hakkinen *et al.*, 1986; Aagaard *et al.*, 2002a; Suetta *et al.*, 2004; Bojsen-Moller *et al.*, 2005; Caserotti *et al.*, 2008; Thorlund *et al.*, 2008; Dewhurst *et al.*, 2010; Thompson *et al.*, 2012, 2013; Waugh *et al.*, 2013, 2014; Haff *et al.*, 2015), absolute RFD was derived as the average slope of the force–time curve over specified time intervals of 0–30, 0–50, 0–100, 0–200 and even 0–400 ms relative to the onset of contraction, whereas MRFD typically has been quantified as the peak slope of the force–time curve (Gorostiaga *et al.*, 1999; Sahaly *et al.*, 2001; Aagaard *et al.*, 2002a; Mirkov *et al.*, 2004; Suetta *et al.*, 2004; Bojsen-Moller *et al.*, 2005; Korhonen *et al.*, 2006; Holtermann *et al.*, 2007; Hartmann *et al.*, 2009; Almosnino *et al.*, 2010; Thompson *et al.*, 2012, 2013; Haff *et al.*, 2015). RFD has also been calculated as the time taken from the beginning of force production to the attainment of different fractions (e.g. 25%, 50% and 75%) of the maximal voluntary contraction (Hakkinen & Hakkinen, 1991; Sleivert & Wenger, 1994; Gorostiaga *et al.*, 1999; Newton *et al.*, 2002), as the time interval elapsed between achieving different fractions (e.g. 10% and 90%) of the maximal force (Viitasalo *et al.*, 1981; Matavulj *et al.*, 2001; Mirkov *et al.*, 2004) and as the time taken to increase the force from contraction onset to a given level (e.g. 100, 250, 500, 750, 1000, 1500 and 2000 N) (Hakkinen *et al.*, 1985; Baker *et al.*, 1994; Korhonen *et al.*, 2006). Other variables, including the force exerted at different time periods (e.g. 30, 100 or 500 ms) after the contraction initiation (Wilson *et al.*, 1995; Izquierdo *et al.*, 1999a; Mirkov *et al.*, 2004; Hannah *et al.*, 2012; Tillin *et al.*, 2013; Tillin & Folland, 2014; Haff *et al.*, 2015), the time from force onset to the instant of reaching MRFD (Kawamori *et al.*, 2006; de Ruiter *et al.*, 2007, 2010; Thorlund *et al.*, 2008) and the time needed to reach one-sixth, one-half and two-thirds of the maximal voluntary force, have also been calculated to evaluate the rapid force production. Other quantification methods for RFD include calculating the force impulse (force or torque integrated with respect to time) in specific time intervals, such as 0–30, 0–100 or 0–250 ms relative to force onset (Wilson *et al.*, 1995; Aagaard *et al.*, 2002a; Suetta *et al.*, 2004; West *et al.*, 2011; Thompson *et al.*, 2012; Bazyler *et al.*, 2015), or specified as the ratio between the peak force and the time it took to reach such peak force (Nuzzo *et al.*, 2008; Haff *et al.*, 2015).

In dynamic RFD assessments using CMJ testing, contractile RFD has been determined as the difference in the vertical force–time record from the point of peak ground reaction force (GRF) minus GRF recorded at the onset of the eccentric deceleration (upward acceleration) phase, divided by time (Ebben *et al.*, 2007, 2008; Thorlund *et al.*, 2008). In other studies, average RFD was calculated as the maximum vertical GRF achieved over the force–time curve during the CMJ jump divided by the time taken to achieve such maximum force (Dowling & Vamos, 1993; Ugrinowitsch *et al.*, 2007; Moir

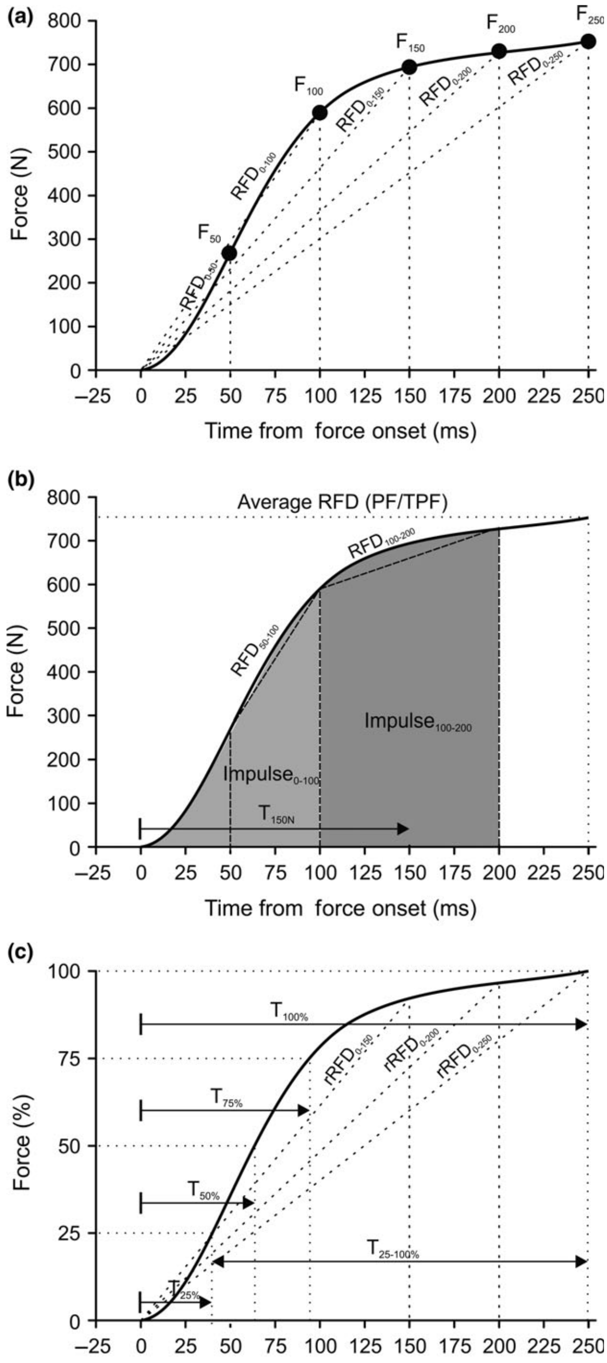

Figure 4 Common methods used to quantify RFD of the force–time curve. (a) The average tangential slope of the force–time curve obtained over different time intervals (0–50, 0–100, 0–150, 0–200 and 0–250 ms) relative to the onset of contraction (t = 0 s) and force exerted at specific time points (F50, F100, etc.); (b) Time taken to increase the force from contraction onset to a given level (e.g. 150 N), sequential RFD and sequential impulse both assessed over consecutive time periods (0–50, 50–100, 100–150, 150–200 ms); (c) Time taken from the beginning of force production to the attainment of different fractions of the maximal force (e.g. 25%, 50% 75% and 100%), time interval elapsed between achieving different fractions (e.g. 25% and 100%) of the maximal force and the average tangential slope of the force–time curve obtained when normalized relative to peak force over different time intervals (0–50, 0–100, 0–150, 0–200 and 0–250 ms) relative to the onset of contraction (normalized RFD).

et al., 2009; McLellan *et al.*, 2011), whereas MRFD also has been calculated as the peak slope of the GRF force–time curve from the minimum force value achieved during the movement to the maximum force achieved in the jump (Wilson *et al.*, 1995; de Ruiter *et al.*, 2006; Moir *et al.*, 2009; McLellan *et al.*, 2011; Jakobsen *et al.*, 2012; Laffaye & Wagner, 2013; Laffaye *et al.*, 2014). In all the above-mentioned studies, RFD was quantified during the eccentric phase (downward movement) of the jump (Hori *et al.*, 2009). This seems to be the only phase during the CMJ in which it makes sense to measure RFD, as the applied force during the initial part of the concentric phase tends to plateau or be lower than the force produced during the final part of the eccentric phase. Consequently, the change in force in relation to time (Δforce/Δtime = RFD) is expected to be very small or even negative in the concentric take-off phase (upward movement), which may result in very low RFD values. Despite this, several studies have managed to calculate RFD during the concentric phase of the jump (Cormie *et al.*, 2008; Moir *et al.*, 2009).

As for absolute RFD, contrasting approaches exist to quantify the relative or normalized RFD. Thus, normalized RFD has been determined as: (i) the slope of the force–time curve when the force values were normalized relative to maximum force (expressed as %MF s^{-1}) (Aagaard *et al.*, 2002a; Holtermann *et al.*, 2007; Thorlund *et al.*, 2008; Andersen *et al.*, 2010; Thompson *et al.*, 2012); (ii) absolute MRFD divided by the maximum force (MRFD MF^{-1}) (Waugh *et al.*, 2013, 2014), body mass (MRFD BM^{-1}) (Andersen *et al.*, 2007) or muscle CSA (MRFD CSA^{-1}) (Suetta *et al.*, 2004); and (iii) rate of increment in relative force from the onset of contraction to the level of one-sixth, one-half and two-thirds of maximum isometric force, respectively (Aagaard *et al.*, 2002a; Suetta *et al.*, 2004; Blazevich *et al.*, 2008; Thorlund *et al.*, 2008).

Determining relative RFD is of paramount importance because absolute RFD appears to be influenced by the peak force capacity of the muscles (Ryushi *et al.*, 1988; Andersen *et al.*, 2005). In fact, although RFD and maximum dynamic or isometric force are considered as independent variables that express different functional and neuromuscular abilities (de Ruiter *et al.*, 2008), numerous studies have observed concurrent changes in both strength parameters with ageing (Bemben *et al.*, 1991; Paasuke *et al.*, 2001b), resistance training (Hakkinen *et al.*, 1985; Behm & Sale, 1993; Van Cutsem *et al.*, 1998; Rich & Cafarelli, 2000; Aagaard *et al.*, 2002a) or acute muscle fatigue (Thorlund *et al.*, 2009; Molina & Denadai, 2012; Penailillo *et al.*, 2015). In addition, a positive association exists between the RFD and maximal force (Demura *et al.*, 2003; Mirkov *et al.*, 2004), especially for RFD recorded in the later phase of rising muscle force (Andersen & Aagaard, 2006). However, Holtermann *et al.* (2007) indicated that this association between RFD and maximal force can be either causal or mediated by a third factor ('confounder factor'). In line with these findings (Holtermann et al., 2007), only a weak relationship between maximum isometric force and absolute RFD has been reported to exist (Jaric et al., 1989; Murphy et al., 1995; Beckham et al., 2013). Thus, absolute RFD and relative RFD seem to reflect different neuromuscular qualities. While the training-induced change in absolute RFD contains information about both the *force* and *time* aspects of the maximal voluntary contraction, data on relative RFD yield information about the *time* aspect of the force–time slope alone and are useful for studying physiological mechanisms influencing the expression of RFD independently of maximal generated force or anatomical muscle size (Holtermann et al., 2007; Blazevich et al., 2008; Waugh et al., 2014). Notably, relative RFD allows different muscles and/or individuals with different muscle development (e.g. sedentary subjects versus strength-trained athletes; children versus adults; men versus women) to be compared. However, the normalization of RFD related to maximum force or torque also has potential limitations: (i) relative RFD may be increased when peak force is not truly maximal (Sahaly et al., 2001); and (ii) relative RFD represents an incomplete stand-alone measure when evaluating whether an individual's rate of applied force has improved after a period of training. For example, RFD expressed relative to maximum force (RFD/Fmax) can be increased without this necessarily involving a performance improvement, as functional capacity (i.e. movement speed, acceleration) is exclusively governed by absolute and not normalized muscle strength/RFD parameters. Also, an identical relative decrease in both maximum force and RFD would result in unaltered relative RFD, but still result in a significant performance loss in high-speed muscle actions. Therefore, for a better understanding of the characteristics of an individual and/or the effects of a training programme in relation to the ability to apply force rapidly, data on relative RFD value should always be accompanied by concurrent data on absolute RFD.

Type of muscle contraction

Contractile RFD can be measured in both isometric and dynamic muscle actions. Most studies have assessed RFD using isometric test contractions including leg extension (Hakkinen et al., 2003; Korhonen et al., 2006), knee flexion or extension (Izquierdo et al., 1999b; Aagaard et al., 2002a; Andersen et al., 2010; Zebis et al., 2011), plantar flexion and extension (Waugh et al., 2013, 2014), squat (Young & Bilby, 1993; Tillin et al., 2013), mid-thigh clean pull (Haff et al., 2005), bench press (Pryor et al., 1994; Murphy & Wilson, 1996), elbow flexion and extension (Mirkov et al., 2004; Ingebrigtsen et al., 2009), and even during static neck flexion and extension (Almosnino et al., 2010). Proponents for the use of isometric tests to assess contractile RFD argue that there is a high level of control in the measurement (force–length and force–velocity properties remain identical between repeated test sessions), and therefore, these tests have generally shown high reliability in both single and multijoint test protocols (Abernethy et al., 1995; Wilson & Murphy, 1996; Buckthorpe et al., 2012). However, isometric muscle testing may have potential limitations (Abernethy et al., 1995; Wilson & Murphy, 1996) including (i) potentially low external validity of isometric assessment, which might negatively affect the relationship to athletic performance; (ii) examining a highly restricted region of the force–length relationship dictated by the choice of joint angle(s) used in the isometric assessment; and (iii) some isometric testing positions might be uncomfortable and potentially injurious.

Dynamic RFD has typically been assessed using different joint movements such as isolated knee extension (Dewhurst et al., 2010; Molina & Denadai, 2012), hip extension (LaRoche et al., 2008), squat (Sleivert & Taingahue, 2004), elbow flexion and extension (Mirkov et al., 2004; Adamson et al., 2008; Ingebrigtsen et al., 2009), bench press (Pryor et al., 1994; Wilson et al., 1996), weightlifting movements (Haff et al., 2005; Kilduff et al., 2007; Comfort, 2013) and different jump tests either with CMJ and DJ or without SJ stretch-shortening cycle involvement (SSC) (Haff et al., 1997; Marcora & Miller, 2000; Kawamori et al., 2005; McLellan et al., 2011; Jakobsen et al., 2012). Although only few studies have compared RFD values across different contraction modes (Pryor et al., 1994; Wilson et al., 1995; Haff et al., 1997, 2005; Tillin et al., 2012a), it seems that different muscle actions (concentric, eccentric, isometric or SSC) produce large variations in RFD. In general, the results of these studies indicate that absolute RFD values are lower during isometric compared to concentric (Pryor et al., 1994; Wilson et al., 1995; Haff et al., 1997, 2005) or eccentric (Pryor et al., 1994) test contractions, whereas the differences between absolute concentric and eccentric RFD remain less clear (Pryor et al., 1994). In addition, Tillin et al. (2012a) showed that differences between concentric and isometric RFD were greater after normalization of RFD values to the maximal force obtained in each contraction mode. Collectively, these results demonstrate that the mode of muscle contraction has a profound effect on the RFD values obtained and consequently is likely to influence the potential relationship between isolated RFD capacity and dynamic movement performance (Wilson et al., 1995).

Measuring devices

Many types of devices have been used to measure RFD. A large number of studies have employed instrumented force plate technology (Wilson et al., 1993; Young & Bilby, 1993; Haff et al., 1997, 2005; Izquierdo et al., 1999a; Kawamori et al., 2005, 2006; Kilduff et al., 2007; Ugrinowitsch et al., 2007; McLellan et al., 2011; Jakobsen et al., 2012; Laffaye & Wagner, 2013; Laffaye et al., 2014) or electromechanical/isokinetic dynamometers (Izquierdo et al., 1999b; Aagaard et al., 2002a; Hakkinen et al., 2003; Suetta et al., 2004; Bojsen-Moller et al., 2005; Ricard et al., 2005; Thorlund et al., 2008;

Andersen et al., 2010) to measure the force–time curve in different types of muscle actions. In addition, several studies have used a linear position transducer (Bemben et al., 1991; Chiu et al., 2004; Gonzalez-Badillo & Marques, 2010; Hansen et al., 2011) or velocity transducer (Fernandez-Del-Olmo et al., 2014; Balsalobre-Fernandez et al., 2015; Hernandez-Davo et al., 2015). However, some controversy arise for RFD when measured using linear position or velocity transducer technology, because these devices do not directly measure the applied force. In fact, the force values exhibited by linear position encoders or velocity transducers are derived from the load mass and the acceleration of the load. This means that these instruments do not allow measurement of RFD in the static phase of dynamic contraction (before the displacement begins), where the MRFD is achieved (Jimenez-Reyes et al., 2016). Therefore, these devices should not be used as a standalone measure of RFD. To our best knowledge, only a single previous study (Hansen et al., 2011) has compared RFD obtained at different time intervals (0–100, 0–200, 0–300 ms) during CMJ testing by two different test devices: force plates and linear position transducers. The results of this study (Hansen et al., 2011) showed greater values for all RFD variables analysed when using the linear position transducer compared to force plate. Differences in RFD values obtained between force platform and linear position transducer could be due to the signal processing (mainly noise filtering procedures), which likely is different for each different device. Therefore, as RFD is measured from the force–time or torque–time curve, it is recommended to use force platforms for calculating dynamic RFD values, as these devices allow to directly and instantaneously measure the force applied throughout the joint of BCM range of motion.

Joint angle specificity during isometric activations

As noted above, quantification of RFD has typically been performed by isometric testing (Abernethy et al., 1995; Wilson & Murphy, 1996; Haff et al., 1997, 2015; Aagaard et al., 2002a; Suetta et al., 2004; Bojsen-Moller et al., 2005). Most researchers have conducted isometric measures at a single joint angle in the movement range, often choosing the joint angle that produce maximal torque generation. However, it is has been well documented that the isometric force-producing capabilities of muscles (RFD and peak force) vary as a function of joint angle (Sale, 1991; Murphy et al., 1995; Wilson & Murphy, 1996). Studies performed by Murphy & Wilson (1996) and Murphy et al. (1995) examined RFD produced during isometric bench press exercise at elbow angles of 120° compared to 90° and noted that RFD at each static joint angle was significantly different, with RFD being greater in the 120° position. In addition, these authors (Murphy et al., 1995) noted that the correlation coefficient between RFD obtained at elbow angles of 90° and 120° was smaller than what may have been expected ($r = 0{\cdot}62$; $P<0{\cdot}05$). Similarly, a recent study (Bazyler et al., 2015) showed a low shared variance between RFD at 90° and 120° obtained during static squat exercises ($R^2 = 0{\cdot}20$). Thus, it appears that when an identical isometric movement is changed by only 30° of joint angle, the magnitudes of the changes in RFD are subject to large individual variations (Murphy et al., 1995; Bazyler et al., 2015). In line with these results, substantial differences in contractile RFD appear to exist when measured at different joint angles during isometric leg extension or knee extension (Marcora & Miller, 2000; de Ruiter et al., 2004; Tillin et al., 2012a). However, such differences may at least in part also be affected by the distinct methods used to quantify RFD. For example, Tillin et al. (2012a) compared the absolute and normalized torques measured at 25-ms intervals up to 150 ms after torque onset during voluntary isometric knee extensions at knee joint angles of 101° and 155°. Absolute knee extensor force at joint angle of 101° was greater than absolute torque at knee joint angle of 155° during all time intervals analysed except at 25 ms (Tillin et al., 2012a), whereas normalized force values were similar between both conditions for all time intervals analysed (Tillin et al., 2012a). On the other hand, de Ruiter et al. (2004) found that the RFD obtained at 120° knee flexion enabled researchers to distinguish between good and poor jumpers, whereas RFD measurements obtained at 90° did not. The results of these studies strongly underline that the joint angle at which isometric testing takes place should not be chosen arbitrarily, given that the relationship between isometric RFD data as well as between the isometric tests and performance varies substantially as a function of joint angle. In this context, Murphy et al. (1995) recommended that isometric tests should be performed near or at the joint angle at which peak force is achieved during the movement of interest, whereas other authors (Sale, 1991) recommended that isometric testing should be performed at the specific joint angle that corresponds to the peak of the strength–joint angle (force–length) curve for that particular muscle group, in order to reduce the variability in force output.

Evoked muscle contractions

Fast muscle contractions can be voluntarily performed or electrically induced. Because muscle activation appears to be the most important factor in the rate of force rise at the onset of force production, it is assumed that the maximal voluntary RFD will be slower than electrically evoked RFD because initial muscle activation at the start of force rise is expected to be lower during voluntary contractions. In line with this hypothesis, several studies (Koryak, 1998; de Ruiter et al., 2007, 2010; Tillin et al., 2012b) have shown that RFD appears to be higher and achieved sooner during electrically evoked contractions than in voluntary contractions, which was also accompanied by a lower EMD (Tillin et al., 2010). In contrast, de Ruiter et al. (2004) found that voluntary and electrically induced MRFDs were similar during isometric testing of the knee extensors, regardless of the joint angle examined. However, the same study (de Ruiter et al., 2004) also reported that the time to achieve MRFD was, on average, almost twice as long during voluntary compared with electrically induced contractions. Furthermore, the force–time integral over the first 40 ms was significantly higher for the electrically induced compared with the voluntary fast contractions. As suggested by de Ruiter *et al.* (2004), it likely takes longer before maximal saturating intracellular calcium concentrations are reached during maximal voluntary activation than when using electrical-burst muscle stimulation at 300 Hz. In conclusion, it seems that differences between voluntary and electrically induced RFD are dependent on the specific method used to quantify RFD.

Muscular pretension

Even though some researchers have recommended that testing subjects should maintain a constant tension against the bar prior to beginning an isometric maximal activation (Bazyler *et al.*, 2015), an excessive level of tension has been shown to affect isometric assessment, particularly the magnitude of RFD. Viitasalo (1982) analysed the force–time curve characteristics during isometric knee extension using pretensioning at various submaximal contraction levels. The contractions were performed from seven different pretension levels: 0%, 20%, 30%, 40%, 50%, 60% and 70% of maximal voluntary contraction value. All levels of pretension caused a significant reduction in MRFD, which was more pronounced as initial pretension increased. Similarly, Van Cutsem & Duchateau (2005) showed greater MRFD during isometric plantar flexion when the contraction was performed from a resting condition compared to a ballistic contraction from a sustained contraction (~25% of MVC). It is plausible that during rapid muscle contractions performed from a sustained force level, the average discharge rate of motor units and the percentage of units that exhibit double discharges significantly decrease and that muscle activation is less synchronized (Van Cutsem & Duchateau, 2005). Based upon the findings of these studies (Viitasalo, 1982; Van Cutsem & Duchateau, 2005), it appears that isometric RFD and MVC assessment should be conducted with minimal levels of muscular pretension in order to obtain a more valid measure of the ability to produce force quickly.

Type of instruction

Another problem confronting investigators interested in obtaining accurate estimates of neuromuscular function in humans is choosing an instruction that will elicit a maximal voluntary effort throughout the entire time course of contraction (Christ *et al.*, 1993). This issue has been addressed in several studies (Bemben *et al.*, 1990; Christ *et al.*, 1993; Sahaly *et al.*, 2001, 2003; Holtermann *et al.*, 2007) which have highlighted the importance of instructions on both RFD and peak force. Christ *et al.* (1993) studied the effect of instruction type (*fast* versus *hard*) on isometric RFD obtained for six muscle groups in untrained women aged 25–74 years. In this study, the use of the *fast* instruction yielded greater RFD values (23·6–53·7%) than the *hard* instruction. Similarly, a study performed by Bemben *et al.* (1990) also found that RFD was significantly greater when subjects were instructed to produce the maximal force as *fast* as possible than when they were instructed to produce a very *strong* isometric contraction. More recently, Sahaly *et al.* (2001) analysed the effect of instructions given to test subjects upon RFD measured using different test contraction modalities and muscle groups, respectively, while also differing with regard to muscle mass (elbow flexor, unilateral and bilateral leg extensors), habitual usage (arms compared to legs, take-off leg compared to lead leg) and biomechanics (unilateral compared to bilateral leg extension). The results of their study indicated that higher RFD values were obtained when using instructions of a *fast contraction* instead of the conventional instruction of *strong and fast* muscle actions. Similar results have also been reported by other authors (Sahaly *et al.*, 2003; Holtermann *et al.*, 2007). Furthermore, Holtermann *et al.* (2007) found that the effect of instruction on RFD depended on the time interval used to measure this variable. Thus, the *fast* instruction showed increased absolute RFD and relative RFD in the early contraction phase only (the first 100 ms), whereas the *strong and fast* instructions yielded the highest relative RFD (i.e. RFD normalized to MVC) in the later phase of rising muscle force (after 200 ms). These observations have major implications for the assessment of RFD and suggest that the test leader must be clear about the instructions given and the muscular quality to be assessed.

Surface stability

In the recent years, training on unstable surfaces has emerged as a novel training method to improve physical performance (Gruber & Gollhofer, 2004; Gruber *et al.*, 2007; Romero-Franco *et al.*, 2012). Descriptive studies comparing acute peak force and muscle activity (EMG) on stable and unstable surface suggest that both variables are reduced when using unstable compared to stable surfaces (Behm *et al.*, 2002; Kohler *et al.*, 2010; Saeterbakken & Fimland, 2013). With respect to RFD, McBride *et al.* (2006) reported that the RFD during isometric squats was significantly lower (40·5%) in the unstable versus stable condition. Furthermore, EMG activity for the quadriceps muscles was also significantly lower (~35%) in unstable compared to stable conditions (McBride *et al.*, 2006). Therefore, the use of resistance exercises on unstable surfaces may be questioned when the aim of the training is improve the ability to generate high RFD.

Testing position

Muscle activation and maximal strength studies are typically conducted using seated or standing positions. However, there are a number of scenarios in which an individual may have to perform muscle contractions under supine or inverted conditions (Paddock & Behm, 2009). Several central and peripheral factors that regulate neuromuscular function are altered under inverted conditions. Thus, various studies have compared RFD measured in both upper- (Hearn et al., 2009) and lower-limb muscles (Paddock & Behm, 2009) in neutral and inverted positions. These studies showed significant decreases in the RFD and muscle activity of selected muscles in inverted positions. This effect of body position on neuromechanical performance has been hypothesized to be attributable to a blunted sympathetic response, increased cerebral blood pooling, decreased afferent reflex activity and/or reduced central activation of the involved muscle when testing is performed in a inverted or lying position compared to upright positions (Maffiuletti & Lepers, 2003; Hearn et al., 2009; Paddock & Behm, 2009). To facilitate a maximal voluntary effort, therefore, RFD testing is recommended to occur in upright, non-supine upper body positions.

Temperature effects

The effect of muscle temperature on neuromuscular performance has been widely studied in both young and older individuals (Bennett, 1984; Sargeant, 1987; Holewijn & Heus, 1992; de Ruiter & De Haan, 2000; Cheung & Sleivert, 2004; Gray et al., 2006). It appears that muscle heating produces an increment in peak force (Gray et al., 2006), whereas a lowering muscle temperature has the opposite effect (Cheung & Sleivert, 2004). However, the effect of muscle temperature on RFD is less clear. Some studies have noted significant increments in RFD after muscle heating and substantial declines following muscle cooling (de Ruiter et al., 1999; de Ruiter & De Haan, 2000). Other studies have reported significant decreases in RFD in cooling conditions, whereas passive heating conditions did not affect RFD (Holewijn & Heus, 1992). The observed decline in RFD at lower muscle temperatures likely was caused by reduced neural activation in cooling conditions compared to heating or control conditions (Holewijn & Heus, 1992; Dewhurst et al., 2010). In contrast, several studies have showed no changes in RFD when muscle temperature is altered. For example, Ranatunga et al. (1987) analysed the changes in RFD relative to peak force in human first dorsal interosseous muscle over a wide range of muscle temperatures (from 12 to 35°C). The results of this study indicated that relative RFD was largely independent of muscle temperature. Similarly, Dewhurst et al. (2010) investigated the effect of altered muscle temperature on RFD measured during maximal isometric knee extensor and flexor contractions in young and older women. Tests were performed at three different muscle temperatures: control (~34°C), cold (~30°C) and warm (~38°C). Neither cooling nor warming had a measurable effect on RFD in either group for any of the time intervals (0–30, 0–50, 0–100, 0–200 ms). The lack of relationship between changes in muscle temperature and changes in RFD reported in some of these studies (Ranatunga et al., 1987; Dewhurst et al., 2010) might be caused by a very small within-protocol difference in muscle temperature. Therefore, the effect of muscle temperature on RFD remains at least partly unclear, and more studies are needed to determine the specific influence of muscle temperature on neuromuscular function.

Other methodological factors affecting the measurement of rate of force development

In addition to the aspects mentioned above, recent studies have indicated that the measurement of RFD may also be significantly influenced by less obvious methodological factors such as jaw clenching (Ebben et al., 2008), knowledge of the load to be lifted (Hernandez-Davo et al., 2015) and the physical environment (aquatic versus dry land) in which the test is conducted (Triplett et al., 2009). However, given the limited amount of data addressing the influence of these factors on the measurement of RFD, further studies seem needed to more consistently map out the effect on RFD obtained in standardized and controlled test conditions.

Summary

As described above, RFD measurements can be influenced by a multitude of methodological factors (Fig. 3), including measurement devices, types of muscle contraction, methods and instruction used to evoke contraction, and design of postrecording analysis algorithms. These factors, together with the substantial degree of interindividual variability in RFD (Folland et al., 2014) and wide variety of methods that can be employed to evaluate RFD, seem to hinder a clear and uniform definition of the RFD concept. In result, comparisons between studies often are difficult to perform in a meaningful way. This support the notion forwarded by Andersen et al. (2010) that the contrasting reports on the magnitude and direction of training-induced plasticity in contractile RFD may not only reflect the effectiveness (or lack hereof) of the different training protocols used, but also might be influenced by the specific assessment method(s) employed.

Conclusions

The main findings of the present review were that (i) RFD can be objectively measured from the force–time or torque–time curve obtained during isometric or dynamic muscle contractions; (ii) RFD evaluated over different time intervals (0–300 ms) following the onset of contraction is of significant interest to scientists and health professionals because it provides insight into the physical condition of patients and athletes and may also provide important information about the effect(s) of training and rehabilitation; (iii) the influence of neural and intrinsic contractile properties on RFD changes throughout the rising force–time curve; thus, early phase of RFD (<100 ms relative to contraction onset) is mainly influenced by neural drive and intrinsic muscle properties, whereas late-phase RFD (≥100 ms) may be more closely related to adaptive mechanisms that promote gains in maximal muscle strength; and (iv) the contrasting results in contractile RFD reported in different studies may at least in part be caused by varying methods used to determine RFD.

Practical applications

RFD is considered an important factor for the performance of motor tasks where the time available to develop force is limited. As such, this variable has wide-ranging implications for athletic performance, tasks of daily life or prevention of falls and musculoskeletal injuries. The evaluation and quantification of changes in RFD with exercise, inactivity or disease provide insight into the physical condition of patients, athletes and healthy old adults and may also provide important information about the effect(s) of long-term training in these populations. Therefore, increased knowledge of the main physiological and methodological factors affecting RFD appears of significant interest to scientists including strength and conditioning researchers and physical conditioning coaches in order to design more sensitive and specific testing protocols and to obtain more accurate and valid measures of RFD. Equally important, increased knowledge of the main physiological factors influencing rapid force production is also of strong importance as providing the background for developing training and rehabilitation programmes aimed at effectively improving these factors in athletes, elderly individuals and patients with a variety of pathologies.

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Conflict of interest

The authors declare that they have no competing interests.

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